From: NILanjan Datta <nil.wid.frnds@gmail.com>
Sent: Thursday, April 25, 2019 11:33 AM
To: lwc-forum
Cc: mridul.nandi@gmail.com; ashwin.jha1991@gmail.com; lightweight-crypto
Subject: OFFICIAL COMMENT: Forgery against SIV-Rijndael256-AEAD and SIV-TEM-PHOTON-AEAD

Dear all,

We have found forgery against SIV-Rijndael256-AEAD and SIV-TEM-PHOTON-AEAD exploiting improper associated data processing.

If the message length is less than 128 bits, two queries with same padded associated data (one with full block and the other with partial) generates same (ciphertext, tag) pair. Formally, the following forging attack can be mounted on SIV-Rijndael256-AEAD:

1. Step 1: Construct A (|A|=256) and A' (|A'| < 256) such that pad(A) = pad(A').
2. Query (N,A,M), with |M| < 128. Let the (ciphertext, tag) pair be (C,T).
3. Forge with (N,A',C,T).

For SIV-TEM-PHOTON-AEAD, take length of A to be 384 and A' to be less than 384 bit and follow the same attack. We have verified both these attacks using the corresponding reference implementations.

Use of different tweaks (based on full / partial) during the final associated data block processing is a solution to this attack.

Thanks and regards,
Nilanjan Datta, Ashwin Jha and Mridul Nandi,
Indian Statistical Institute, Kolkata

From: Tetsu Iwata <tetsu.iwata@gmail.com> on behalf of Tetsu Iwata <iwata@cse.nagoya-u.ac.jp>
Sent: Friday, April 26, 2019 12:53 AM
To: NILanjan Datta; lwc-forum
Cc: mridul.nandi@gmail.com; ashwin.jha1991@gmail.com; lightweight-crypto
Subject: OFFICIAL COMMENT: Forgery against SIV-Rijndael256-AEAD and SIV-TEM-PHOTON-AEAD

Dear all,

We thank Nilanjan Datta, Ashwin Jha and Mridul Nandi for spotting the issue of the domain separation. The analysis is correct, and we will update the domain separation so that it works correctly, and will send the revision to the forum.

Best regards,

SIV-Rijndael256 and SIV-TEM-PHOTON team

On 2019/04/26 0:32, NILanjan Datta wrote:
> Dear all,
>
> We have found forgery against SIV-Rijndael256-AEAD and
> SIV-TEM-PHOTON-AEAD exploiting improper associated data processing.
>
> If the message length is less than 128 bits, two queries with same
> padded associated data (one with full block and the other with
> partial) generates same (ciphertext, tag) pair. Formally, the
> following forging attack can be mounted on SIV-Rijndael256-AEAD:
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> than 384 bit and follow the same attack. We have verified both these
> attacks using the corresponding reference implementations.
>
> Use of different tweaks (based on full / partial) during the final
> associated data block processing is a solution to this attack.
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> Thanks and regards,
> Nilanjan Datta, Ashwin Jha and Mridul Nandi, Indian Statistical
> Institute, Kolkata
>
> --
> To unsubscribe from this group, send email to
> lwc-forum+unsubscribe@list.nist.gov
> Visit this group at
> https://groups.google.com/a/list.nist.gov/d/forum/lwc-forum
> ---
> You received this message because you are subscribed to the Google
> Groups "lwc-forum" group.

From: Tetsu Iwata <tetsu.iwata@gmail.com> on behalf of Tetsu Iwata <iwata@cse.nagoya-u.ac.jp>
Sent: Tuesday, May 07, 2019 11:34 AM
To: NILanjan Datta; lwc-forum
Cc: mridul.nandi@gmail.com; ashwin.jha1991@gmail.com; lightweight-crypto
Subject: [lwc-forum] OFFICIAL COMMENT: Forgery against SIV-Rijndael256-AEAD and SIV-TEM-PHOTON-AEAD
Attachments: sivrijndael256v1.zip; sivtemphotonv1.zip; 00_siv-tem-photon.pdf; 00_siv-rijndael256.pdf

Dear all,

We have revised the document and the reference code so that the domain separation works correctly.
Change log is in the last page of the document.
We once again thank Nilanjan Datta, Ashwin Jha and Mridul Nandi for spotting the issue.

Best regards,

SIV-Rijndael256 and SIV-TEM-PHOTON team

On 2019/04/26 13:52, Tetsu Iwata wrote:
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>
> We thank Nilanjan Datta, Ashwin Jha and Mridul Nandi for spotting the
> issue of the domain separation. The analysis is correct, and we will
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> SIV-Rijndael256 and SIV-TEM-PHOTON team
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>> than 384 bit and follow the same attack. We have verified both these

Forgery on Qameleon and SIV-TEM-PHOTON and SIV-Rijndael256

N.Datta, A.Jha, M.Nandi

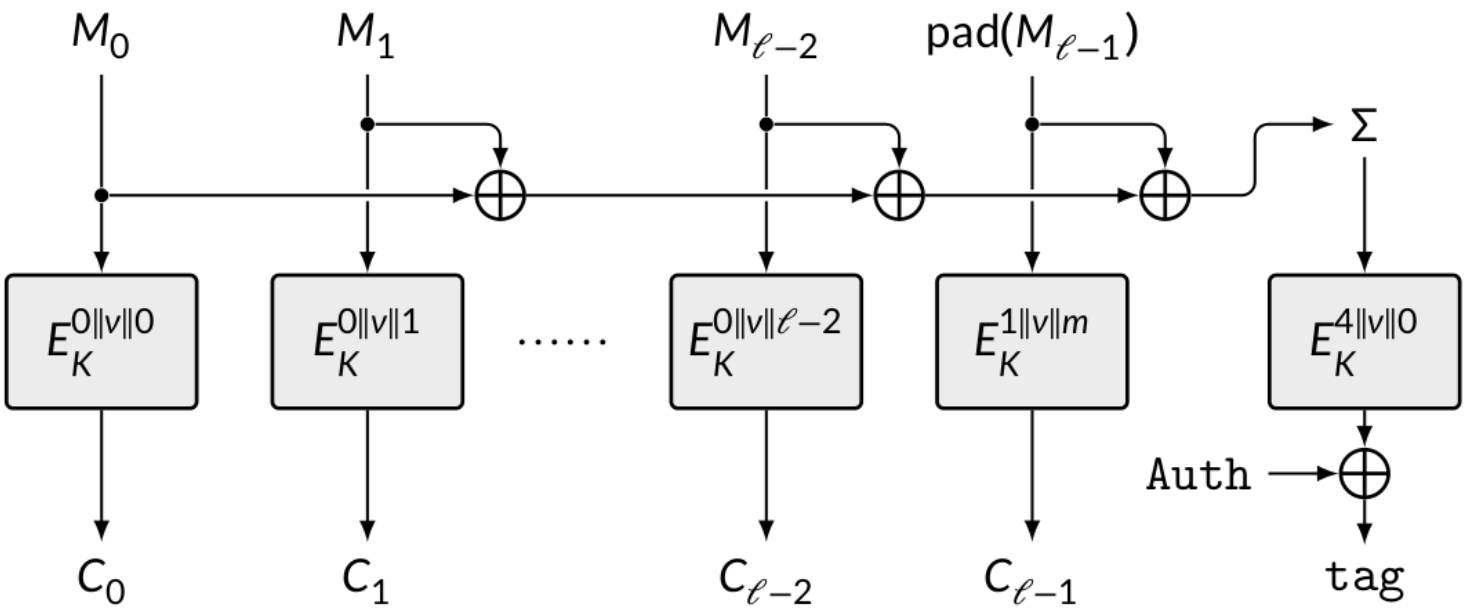
Indian Statistical Institute, Kolkata, India

NIST Lightweight Workshop, 2019

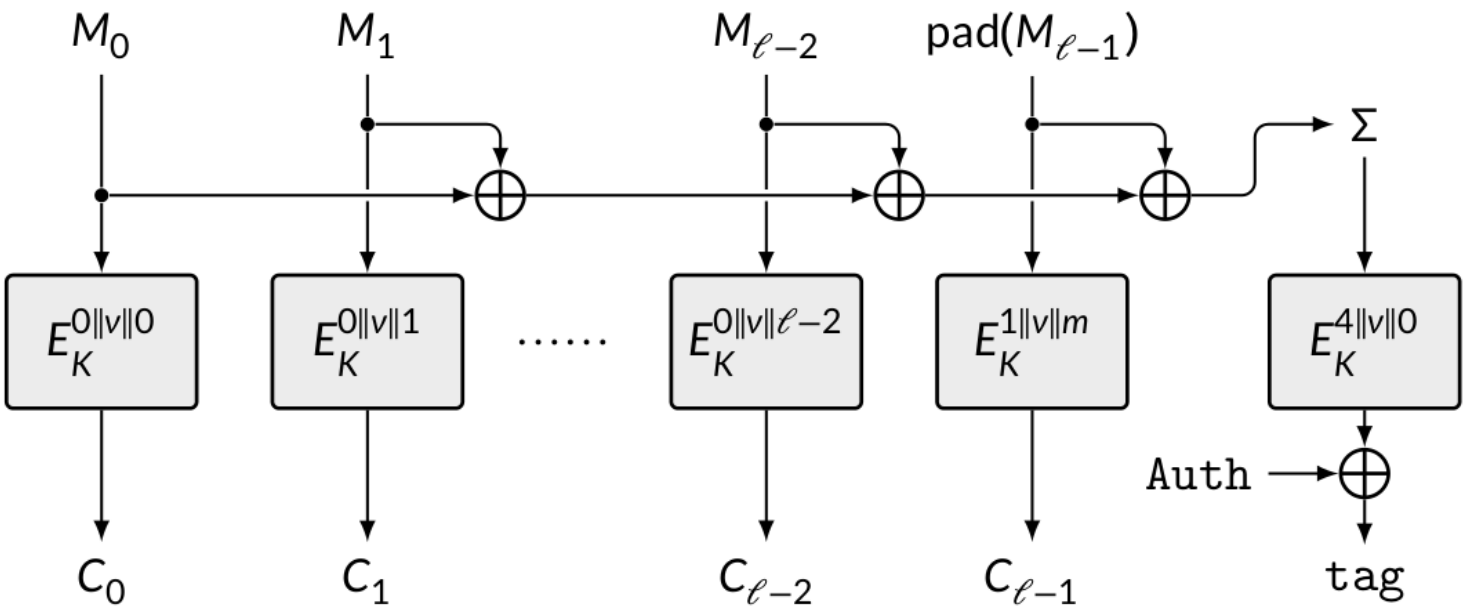
Nov 05, 2019



Qameleon AE Mode



Qameleon AE Mode



- Observation: The message length is not used in the final tweakable block cipher.

Forgery on Qameleon

Forgery Description on Qameleon

- Query $(N,\ A,\ M_1 \| M_1)$ to the encryption oracle. Let $(C_1 \| C_2,\ T)$ be the ciphertext and tag pair.
- Forge with (N, A, ϵ, T), where ϵ denotes empty ciphertext.

Forgery on Qameleon

Forgery Description on Qameleon

- Query $(N,\ A,\ M_1 \| M_1)$ to the encryption oracle. Let $(C_1 \| C_2,\ T)$ be the ciphertext and tag pair.
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Simple Extension

- Take any message $M = M_1 \| \ldots \| M_m$ with $M_1 \oplus \cdots \oplus \text{pad}(M_m) = 0$ and $m < 2^{28}$.
- Query $(N,\ A,\ M)$ to the encryption oracle. Let $(C,\ T)$ be the ciphertext and tag pair.
- Forge with (N, A, ϵ, T), where ϵ denotes empty ciphertext.

How the Forgery Works?

Exploiting improper tweak setting for tag generation

- The AD is same in both the cases.
- The checksum of M matches with the checksum for empty message, i.e. 0.
- The tweak value for tag generation block cipher call is same in both the cases, i.e. , $4\|v\|0$ (since nonce is same and $|M|/128 < 2^{28}$).

- Hence, the forgery succeeds with probability 1.

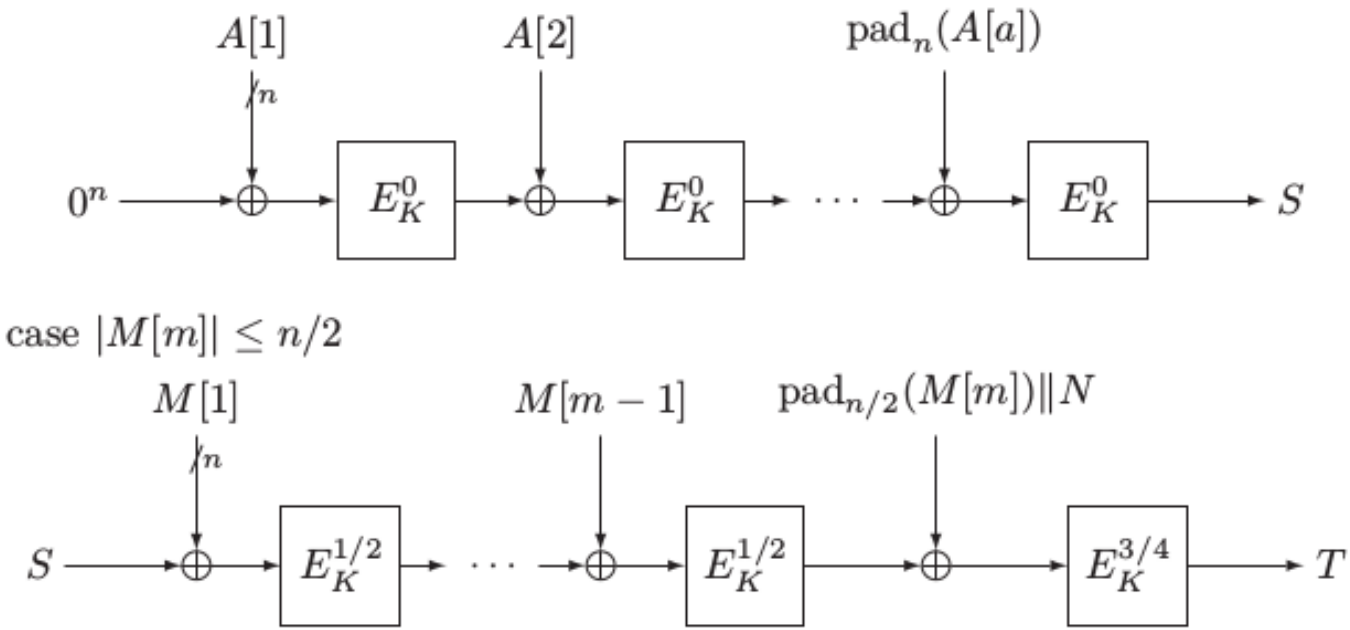
How to Resist the Forgery?

Remark

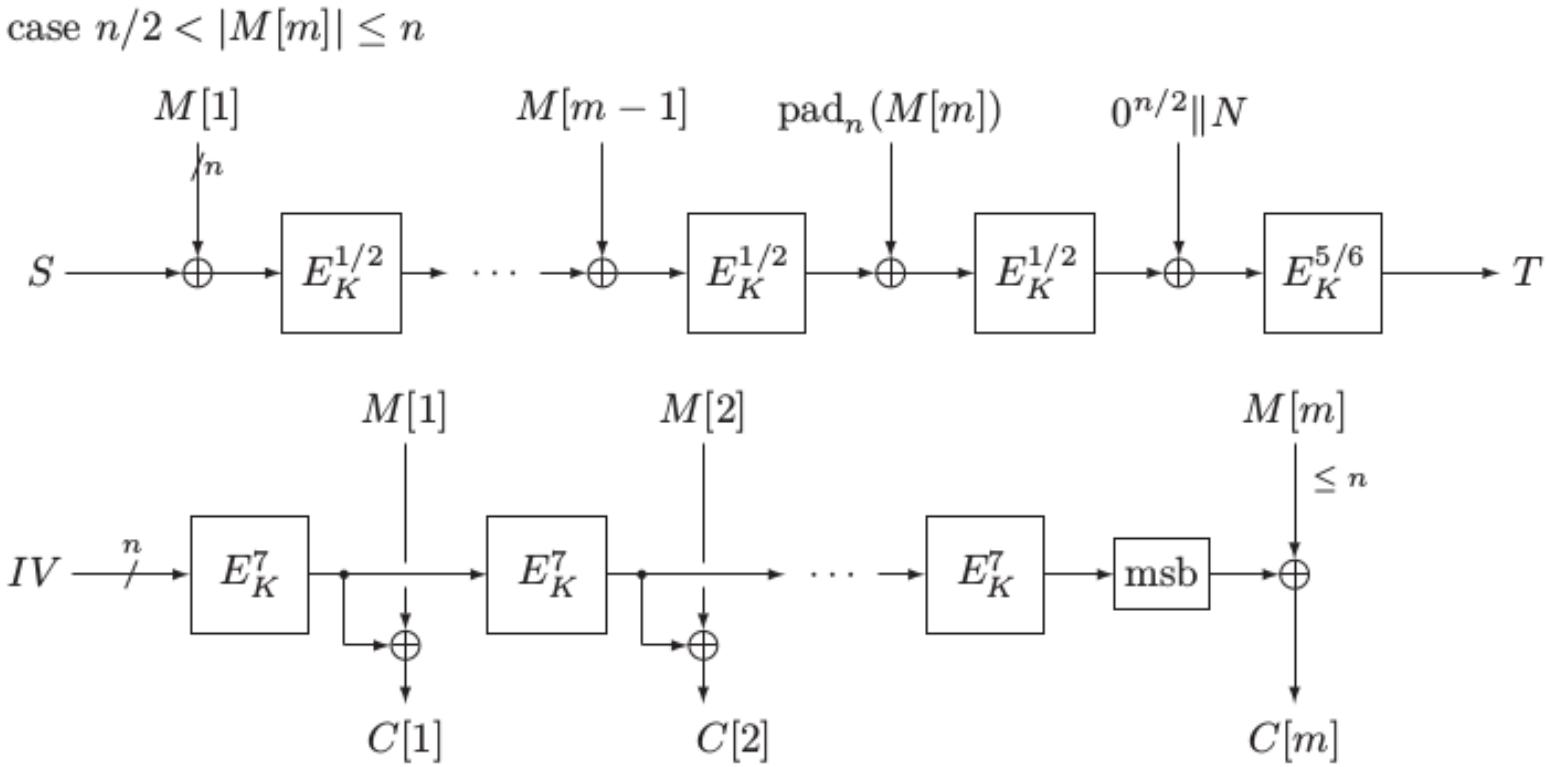
- Forging is possible only with empty message.
- Message length is used for non-empty messages, and hence forging with non-empty message is not possible.

- Use the message length in the tweak of the final tweakable block cipher is a solution to this attack.

SIV-Rijndael256 AE Mode



SIV-Rijndael256 AE Mode



SIV-Rijndael256 AE Mode

Algorithm $\mathcal{F}_K(N, A, M)$

1. $S \leftarrow 0^n$
2. $(A[1], \ldots, A[a]) \xleftarrow{n} A$
3. **if** $|A[a]| < n$ **then** $d \leftarrow 1$ **else** $d \leftarrow 2$
4. $A[a] \leftarrow \mathrm{pad}_n(A[a])$
5. **for** $i = 1$ **to** a **do**
6. $\quad S \leftarrow S \oplus A[i]$
7. $\quad S \leftarrow E_K^0(S)$
8. $(M[1], \ldots, M[m]) \xleftarrow{n} M$
9. **for** $i = 1$ **to** $m - 1$ **do**
10. $\quad S \leftarrow S \oplus M[i]$
11. $\quad S \leftarrow E_K^d(S)$
12. **if** $|M[m]| < n/2$ **then**
13. $\quad S \leftarrow S \oplus (\mathrm{pad}_{n/2}(M[m]) \| N)$
14. $\quad T \leftarrow E_K^3(S)$
15. **if** $|M[m]| = n/2$ **then**
16. $\quad S \leftarrow S \oplus (M[m] \| N)$
17. $\quad T \leftarrow E_K^4(S)$
18. **if** $n/2 < |M[m]| < n$ **then**
19. $\quad S \leftarrow S \oplus (\mathrm{pad}_n(M[m]))$
20. $\quad S \leftarrow E_K^d(S)$
21. $\quad S \leftarrow S \oplus (0^{n/2} \| N)$
22. $\quad T \leftarrow E_K^5(S)$
23. **if** $|M[m]| = n$ **then**
24. $\quad S \leftarrow S \oplus M[m]$
25. $\quad S \leftarrow E_K^d(S)$
26. $\quad S \leftarrow S \oplus (0^{n/2} \| N)$
27. $\quad T \leftarrow E_K^6(S)$
28. **return** T

- Observation: If $|M| \leq n/2$, d is not used in the algorithm, two queries with same padded AD generates same (ciphertext-tag) pair.

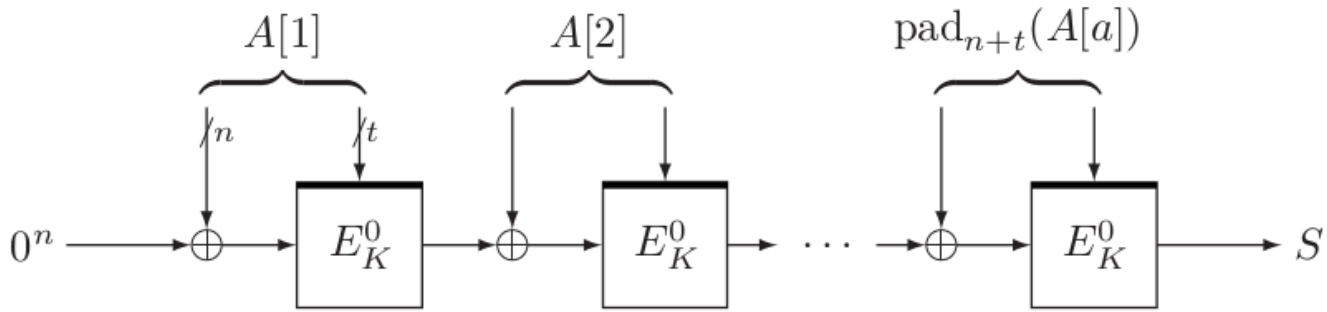
Forgery on SIV-Rijndael256

Forgery Description on SIV-Rijndael256

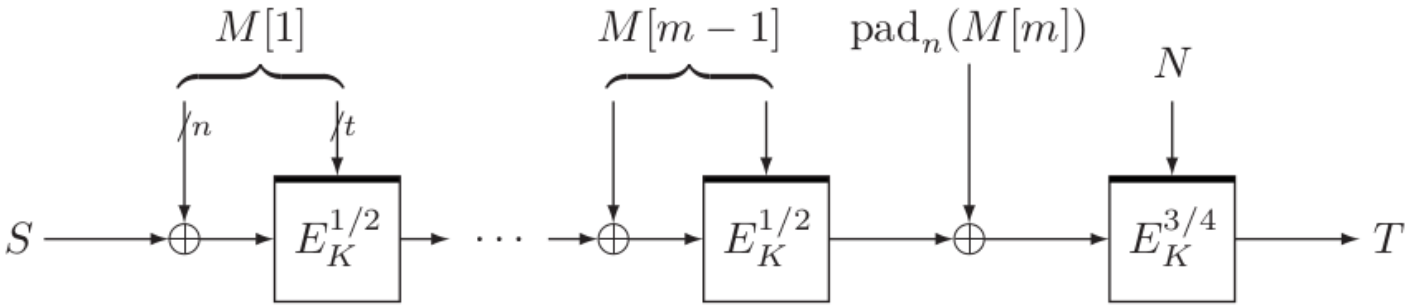
- Construct A ($|A| = 256$) and A' ($|A'| < 256$) such that $\text{pad}(A) = \text{pad}(A')$.
- Query (N, A, M), with $|M| \leq 128$. Let the ciphertext be (C, T).
- Forge with (N, A', C, T).

- The forgery succeeds with probability 1.

SIV-TEM-PHOTON AE Mode



case $|M[m]| \leq n$



SIV-TEM-PHOTON AE Mode

Algorithm $\mathcal{F}_K(N, A, M)$

1. $S \leftarrow 0^n$
2. $(A[1], \ldots, A[a]) \xleftarrow{n+t} A$
3. **if** $|A[a]| < n+t$ **then** $d \leftarrow 1$ **else** $d \leftarrow 2$
4. $A[a] \leftarrow \mathrm{pad}_{n+t}(A[a])$
5. **for** $i = 1$ **to** a **do**
6. $\quad S \leftarrow S \oplus \mathrm{msb}_n(A[i])$
7. $\quad S \leftarrow E_K^{0,\mathrm{lsb}_t(A[i])}(S)$
8. $(M[1], \ldots, M[m]) \xleftarrow{n+t} M$
9. **for** $i = 1$ **to** $m-1$ **do**
10. $\quad S \leftarrow S \oplus \mathrm{msb}_n(M[i])$
11. $\quad S \leftarrow E_K^{d,\mathrm{lsb}_t(M[i])}(S)$
12. **if** $|M[m]| < n$ **then**
13. $\quad S \leftarrow S \oplus \mathrm{pad}_n(M[m])$
14. $\quad T \leftarrow E_K^{3,N}(S)$
15. **if** $|M[m]| = n$ **then**
16. $\quad S \leftarrow S \oplus M[m]$
17. $\quad T \leftarrow E_K^{4,N}(S)$
18. **if** $n < |M[m]| < n+t$ **then**
19. $\quad M[m] \leftarrow \mathrm{pad}_{n+t}(M[m])$
20. $\quad S \leftarrow S \oplus \mathrm{msb}_n(M[m])$
21. $\quad S \leftarrow E_K^{d,\mathrm{lsb}_t(M[m])}(S)$
22. $\quad T \leftarrow E_K^{5,N}(S)$
23. **if** $|M[m]| = n+t$ **then**
24. $\quad S \leftarrow S \oplus \mathrm{msb}_n(M[m])$
25. $\quad S \leftarrow E_K^{d,\mathrm{lsb}_t(M[m])}(S)$
26. $\quad T \leftarrow E_K^{6,N}(S)$
27. **return** T

- Observation: If $|M| \leq n$, d is not used in the algorithm, two queries with same padded AD generates same (ciphertext-tag) pair.

Forgery on SIV-TEM-PHOTON

Forgery Description on SIV-TEM-PHOTON

- Construct A ($|A| = 384$) and A' ($|A'| < 384$) such that $\text{pad}(A) = \text{pad}(A')$.
- Query (N, A, M), with $|M| \leq 256$. Let the ciphertext be (C, T).
- Forge with (N, A', C, T).

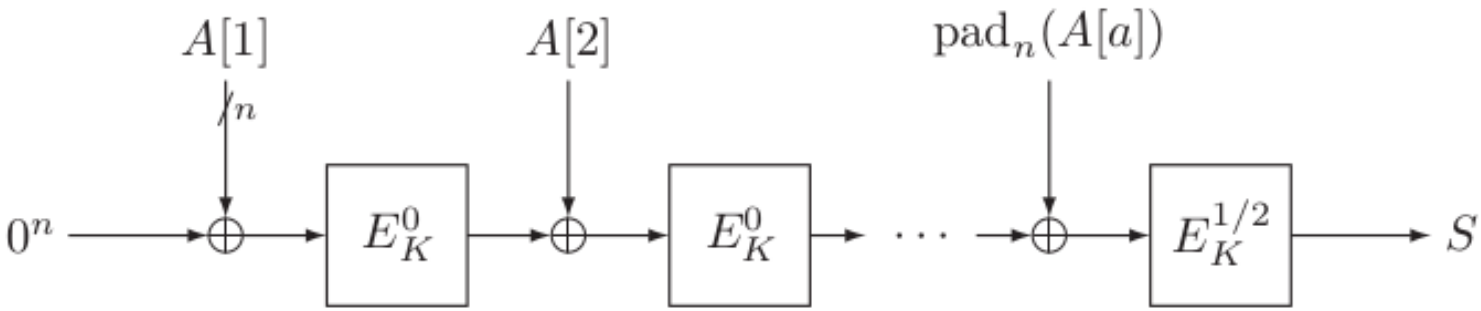
- The forgery succeeds with probability 1.

How to Resist the Forgery?

- Separate the domains for full and partial AD in the AD processing phase.
- Already suggested by the designers in their revised document.

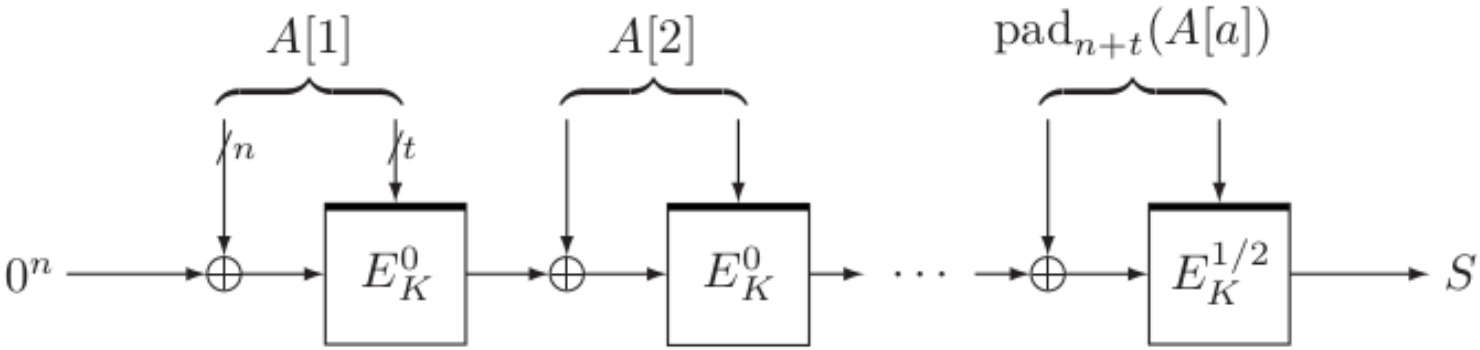
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- Separate the domains for full and partial AD in the AD processing phase.
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Thank You..!! Questions??